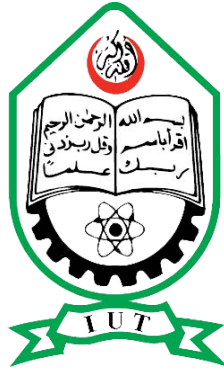


ISLAMIC UNIVERSITY OF TECHNOLOGY



WIRELESS COMMUNICATIONS & NETWORKS

CSE 4615

ALTERNATIVE QUIZ ASSESSMENT

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PART A: CONCEPTUAL QUESTIONS

QUESTION 1 : The IEEE 802.11 standard operates across the Data Link Layer and Physical Layer of the OSI model.

- 1. Explain the division of the Data Link Layer into sublayers and state which sublayer functionality is omitted by IEEE 802.11.**
- 2. Describe the functional roles of the two sublayers within the Physical Layer: Physical Layer Convergence Procedure (PLCP) and Physical Medium Dependent (PMD).**

Answer & Explanation:

1. The IEEE 802.11 Data Link Control (DLC) layer is divided into two sublayers:
 - Logical Link Control (LLC): The LLC sublayer provides higher-level link management, multiplexing, and related functions. IEEE 802.11 does not define a separate LLC implementation; it relies on IEEE 802.2 LLC.
 - Medium Access Control (MAC): The MAC sublayer manages access to the shared wireless medium. It is responsible for MAC framing, addressing, medium access procedures, error detection, and Quality of Service (QoS) mechanisms.
2. The IEEE 802.11 Physical Layer consists of two major sublayers:
 - Physical Layer Convergence Procedure (PLCP): PLCP provides the interface between the MAC layer and the physical transmission system. It takes the MAC data unit and prepares it for transmission by adding the appropriate physical-layer preamble and header, producing a PPDU suitable for transmission.
 - Physical Medium Dependent (PMD): PMD defines the actual method used to transmit and receive information over the wireless medium. It specifies physical signaling, modulation, coding, frequency-related characteristics, and transceiver operations. Different IEEE 802.11 PHY specifications may use technologies such as DSSS, HR-DSSS, or OFDM.

QUESTION 2 : Differentiate between Station Services (SS) and Distribution System Services (DSS) in IEEE 802.11. List two examples of each service type and explain why DSS is not present in an Independent Basic Service Set (IBSS).

Answer & Explanation:

Station Services (SS) are services provided by individual wireless stations. They support operations that are performed by a station regardless of whether it operates in infrastructure or ad-hoc mode.

Examples include:

- MSDU delivery
- Authentication and de-authentication
- Privacy and encryption-related functions

Distribution System Services (DSS) are associated with the Distribution System (DS) and are used to provide communication between stations and to transport frames through an Access Point (AP) when required.

Examples include:

- Association, re-association, and disassociation
- Integration, which supports communication between an IEEE 802.11 network and a non-802.11 LAN through a Portal

DSS is not present in an IBSS because an IBSS operates in ad-hoc mode without an Access Point or centralized Distribution System. Stations communicate directly with one another as peers.

QUESTION 3 : IEEE 802.11 uses Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA). Explain the difference between Physical Carrier Sensing and Virtual Carrier Sensing. Include the following in your explanation:

- 1. Clear Channel Assessment (CCA) threshold.**
- 2. Network Allocation Vector (NAV) mechanism.**

Answer & Explanation:

1. Physical Carrier Sensing:

Physical carrier sensing is performed at the Physical Layer using Clear Channel Assessment (CCA). The wireless interface monitors the received signal and determines whether the medium is busy or idle.

If the detected signal meets the conditions for declaring the medium busy, the station defers transmission. Otherwise, the physical carrier-sensing mechanism may consider the medium idle.

CCA can involve received signal energy detection and/or detection of a valid physical-layer preamble or signal. A threshold is used to determine whether a received signal is sufficiently strong to affect physical carrier sensing.

2. Virtual Carrier Sensing:

Virtual carrier sensing is implemented at the MAC layer using the Network Allocation Vector (NAV).

Frames such as RTS, CTS, and Data contain a Duration field. A station that receives such a frame uses the Duration information to update its NAV timer. While the NAV is greater than zero, the station considers the medium virtually busy and defers transmission.

Therefore:

Physical Carrier Sensing → Determines whether the medium is physically occupied.

Virtual Carrier Sensing → Predicts how long the medium will remain reserved based on information contained in transmitted frames.

QUESTION 4 : Compare IEEE 802.11 unicast and broadcast frame transmission mechanisms with respect to the following four operational parameters:

- 1. RTS/CTS handshake**
- 2. Link-layer acknowledgements and retransmissions**
- 3. Contention window and backoff behavior**
- 4. Data-rate selection**

Answer & Explanation:

Operational Parameter	Unicast Transmission	Broadcast Transmission
RTS/CTS	May use RTS/CTS when applicable	Not used
ACK	ACK is normally expected	No ACK is sent
Retransmission	Failed frames may be retransmitted	No MAC-layer retransmission
Backoff	Random backoff; CW may increase after failure	Contention/backoff is used, but no collision-based retransmission
Data Rate	Can use dynamic rate adaptation	Typically uses a configured basic rate

QUESTION 5 : All stations in an IEEE 802.11 network use the Timing Synchronization Function (TSF) to maintain timing synchronization. Explain how beacon generation and TSF synchronization differ between an Infrastructure Basic Service Set (BSS) and an Independent Basic Service Set (IBSS).

Answer & Explanation:

Infrastructure BSS:

- The Access Point (AP) is responsible for generating Beacon frames.
- The AP attempts to transmit the Beacon at the Target Beacon Transmission Time (TBTT).
- Associated stations receive the Beacon and use its timestamp to synchronize their local TSF timers with the AP.
- The AP therefore acts as the primary timing reference for the BSS.

Independent BSS (IBSS):

- Beacon generation is distributed among the participating stations rather than being controlled by an AP.
- At the Target Beacon Transmission Time, stations may contend to transmit a Beacon.
- A station that successfully obtains the opportunity to transmit sends the Beacon, while other stations cancel their pending Beacon transmission when appropriate.
- A station receiving a Beacon updates its TSF only when the received timestamp represents a later timing value than its current timer.

Thus, an infrastructure BSS obtains synchronization from the AP, whereas an IBSS uses distributed synchronization among participating stations.

PART B: NUMERICAL & ANALYTICAL PROBLEMS

QUESTION 6 : Given the following basic physical parameters for IEEE 802.11a:

$$\text{aSlotTime} = 9 \mu\text{s}$$

$$\text{SIFS} = 16 \mu\text{s}$$

1. Calculate the values of PIFS and DIFS.
2. Explain the fundamental design rule governing why:

$$\text{SIFS} < \text{PIFS} < \text{DIFS}$$

Solution:

1. The Interframe Space relationships are:

$$\text{PIFS} = \text{SIFS} + \text{aSlotTime}$$

$$\text{DIFS} = \text{SIFS} + 2 \times \text{aSlotTime}$$

Therefore:

$$\text{PIFS} = 16 \mu\text{s} + 9 \mu\text{s}$$

$$\text{PIFS} = 25 \mu\text{s}$$

And:

$$\text{DIFS} = 16 \mu\text{s} + 2(9 \mu\text{s})$$

$$\text{DIFS} = 34 \mu\text{s}$$

Therefore:

$$\text{PIFS} = 25 \mu\text{s}$$

$$\text{DIFS} = 34 \mu\text{s}$$

2. The duration of the Interframe Space determines the relative priority of channel access.

SIFS has the shortest duration and therefore provides the highest priority. It is used for immediate responses such as ACK and CTS frames.

PIFS is longer than SIFS but shorter than DIFS. It allows functions associated with centralized channel coordination to obtain access before normal distributed contention.

DIFS is longer than both SIFS and PIFS and is used by stations operating under the Distributed Coordination Function (DCF) before beginning normal contention.

Therefore:

$$\text{SIFS} < \text{PIFS} < \text{DIFS}$$

This ordering prevents ordinary contention traffic from interrupting higher-priority control exchanges.

QUESTION 7 : An IEEE 802.11a station operating under DCF attempts to transmit a frame. The initial contention window parameter is:

$$\text{CW}_{\min} = 16$$

The maximum contention window is:

$$\text{CW}_{\max} = 1024$$

The contention window is represented by values from 0 through CW.

Use the following relationship:

$$\text{CW}[n] = \min(2^{n-1} \times \text{CW}_{\min} - 1, \text{CW}_{\max})$$

Calculate:

- 1. The contention window for attempts $n = 1, 2, 3, 4$, and 8.**
- 2. The maximum possible backoff time for $n = 3$, assuming:**

$$\text{aSlotTime} = 9 \mu\text{s}$$

Solution:

- 1. Attempt 1:**

$$CW_1 = \min(2^0 \times 16 - 1, 1024)$$

$$CW_1 = 15$$

Attempt 2:

$$CW_2 = \min(2^1 \times 16 - 1, 1024)$$

$$CW_2 = 31$$

Attempt 3:

$$CW_3 = \min(2^2 \times 16 - 1, 1024)$$

$$CW_3 = 63$$

Attempt 4:

$$CW_4 = \min(2^3 \times 16 - 1, 1024)$$

$$CW_4 = 127$$

Attempt 8:

$$CW_8 = \min(2^7 \times 16 - 1, 1024)$$

$$CW_8 = \min(2047, 1024)$$

$$CW_8 = 1024$$

Therefore:

Attempt	Contention Window
1	15
2	31
3	63
4	127
8	1024

2. For $n = 3$:

$$CW_3 = 63$$

The maximum possible backoff counter is therefore 63 slots.

$$\text{Maximum Backoff Time} = 63 \times 9 \mu\text{s}$$

$$\text{Maximum Backoff Time} = 567 \mu\text{s}$$

Therefore, the maximum possible backoff time is:

$$567 \mu\text{s}$$

QUESTION 8 : An IEEE 802.11 MAC layer uses an RTS Threshold of:

$R_{\text{th}} = 1000$ bytes

The network uses:

Short Retry Limit = 7

Long Retry Limit = 4

Evaluate the following scenarios:

- 1. Frame A has a size of 800 bytes and repeatedly fails transmission. Which retry counter is used, and after how many unsuccessful attempts is the frame discarded?**
- 2. Frame B has a size of 1500 bytes and repeatedly fails transmission. Which retry counter is used, and after how many unsuccessful attempts is the frame discarded?**
- 3. What happens to the retry counters when an MSDU is successfully acknowledged?**

Solution:

1. Frame A:

Frame A = 800 bytes

RTS Threshold = 1000 bytes

Since:

$800 < 1000$

Frame A is treated as a short frame.

Therefore, the Short Retry Counter (SRC) is used.

The Short Retry Limit is 7, so the frame is discarded when the applicable retry limit is reached after repeated unsuccessful transmission attempts.

Answer:

Short Retry Counter (SRC)

Discard after 7 unsuccessful attempts.

2. Frame B:

Frame B = 1500 bytes

RTS Threshold = 1000 bytes

Since:

$$1500 \geq 1000$$

Frame B is treated as a long frame.

Therefore, the Long Retry Counter (LRC) is used.

The Long Retry Limit is 4.

Answer:

Long Retry Counter (LRC)

Discard after 4 unsuccessful attempts.

3. After a successful transmission and acknowledgement, the corresponding retry state is reset. Under the simplified retry-counter model used in this problem, both SRC and LRC are reset to zero.

QUESTION 9 : IEEE 802.11e Enhanced Distributed Channel Access (EDCA) differentiates channel access among four Access Categories (ACs).

Given:

$$\text{SIFS} = 16 \mu\text{s}$$

$$\text{aSlotTime} = 9 \mu\text{s}$$

Use:

$$\text{AIFS}[\text{AC}] = \text{SIFS} + \text{AIFSN}[\text{AC}] \times \text{aSlotTime}$$

The following EDCA configuration is provided:

Access Category	AIFSN	CWmin	CWmax
AC_VO (Voice)	2	7	15
AC_VI (Video)	3	15	31
AC_BE (Best Effort)	4	31	1023
AC_BK (Background)	7	63	1023

- 1. Calculate the AIFS duration for all four Access Categories.**
- 2. Explain why $AIFSN[AC] \geq 2$ is used for EDCA categories relative to the legacy DIFS definition.**

Solution:

1. AC_VO:

$$AIFS[AC_VO] = 16 + 2 \times 9$$

$$AIFS[AC_VO] = 34 \mu s$$

AC_VI:

$$AIFS[AC_VI] = 16 + 3 \times 9$$

$$AIFS[AC_VI] = 43 \mu s$$

AC_BE:

$$AIFS[AC_BE] = 16 + 4 \times 9$$

$$AIFS[AC_BE] = 52 \mu s$$

AC_BK:

$$\text{AIFS}[\text{AC_BK}] = 16 + 7 \times 9$$

$$\text{AIFS}[\text{AC_BK}] = 79 \mu\text{s}$$

Summary:

Access Category	AIFSN	AIFS
AC_VO	2	34 μs
AC_VI	3	43 μs
AC_BE	4	52 μs
AC_BK	7	79 μs

Therefore:

AC_VO has the shortest AIFS and consequently receives the earliest opportunity to begin contention.

AC_BK has the longest AIFS and consequently has the lowest relative access priority.

2. Legacy DIFS is:

$$\text{DIFS} = \text{SIFS} + 2 \times \text{aSlotTime}$$

$$\text{DIFS} = 16 + 2 \times 9$$

$$\text{DIFS} = 34 \mu\text{s}$$

Therefore, setting AIFSN to at least 2 ensures that the minimum EDCA AIFS is:

$$\text{AIFS}_{\text{min}} = 34 \mu\text{s}$$

This preserves the ordering of access priorities relative to SIFS and PIFS while allowing EDCA to differentiate traffic through larger AIFS values and other EDCA parameters.

PART C: PROTOCOL MECHANICS & SCENARIO ANALYSIS

QUESTION 10 : Two stations, STA-1 and STA-2, contend for a wireless channel in an IEEE 802.11a network.

Given:

$$\text{SIFS} = 16 \mu\text{s}$$

$$\text{aSlotTime} = 9 \mu\text{s}$$

$$\text{DIFS} = 34 \mu\text{s}$$

At $t = 0 \mu\text{s}$, a previous transmission finishes and the medium becomes idle.

Both stations have frames ready to transmit.

STA-1 selects a random backoff counter:

$$r_1 = 3 \text{ slots}$$

STA-2 selects:

$$r_2 = 5 \text{ slots}$$

Trace the exact timeline:

- 1. At what time do both stations complete their DIFS deferral and begin countdown?**
- 2. At what time does STA-1 complete its backoff and initiate transmission?**
- 3. What is the remaining frozen backoff count of STA-2 when STA-1 starts transmitting?**
- 4. Assuming STA-1's complete transmission exchange takes $200 \mu\text{s}$, at what time does STA-2 complete its remaining countdown and initiate transmission after the medium becomes idle again?**

Solution:

1. Both stations must wait for the DIFS interval.

$$\text{DIFS} = 34 \mu\text{s}$$

Therefore:

Countdown begins at $t = 34 \mu\text{s}$.

2. STA-1 has a backoff counter of 3 slots.

Backoff duration:

$$3 \times 9 \mu\text{s} = 27 \mu\text{s}$$

Therefore:

$$t_{\text{transmission}} = 34 \mu\text{s} + 27 \mu\text{s}$$

$$t_{\text{transmission}} = 61 \mu\text{s}$$

STA-1 begins transmission at:

$$t = 61 \mu\text{s}$$

3. STA-2 starts its countdown at the same time and initially has 5 slots.

By the time STA-1 reaches zero, 3 slots have elapsed.

Remaining backoff:

$$5 - 3 = 2 \text{ slots}$$

Therefore, STA-2 freezes its backoff counter at:

2 slots

4. STA-1 begins transmission at:

$$t = 61 \mu\text{s}$$

The complete transmission exchange takes $200 \mu\text{s}$.

Therefore:

$$61 + 200 = 261 \mu\text{s}$$

The medium becomes available again at:

$$t = 261 \mu s$$

STA-2 must wait for another DIFS:

$$261 + 34 = 295 \mu s$$

STA-2 then resumes its frozen backoff of 2 slots:

$$2 \times 9 = 18 \mu s$$

Therefore:

$$t = 295 + 18$$

$$t = 313 \mu s$$

STA-2 initiates transmission at:

$$313 \mu s$$

QUESTION 11 : Consider three stations positioned as follows:

STA-A ↔ AP ↔ STA-B

STA-A and STA-B are outside each other's communication/sensing range, but both are within range of the AP.

STA-A wants to transmit data to the AP using RTS/CTS.

Questions:

- 1. Explain step-by-step why STA-B cannot detect STA-A's physical transmission.**
- 2. Describe how STA-B is prevented from transmitting while STA-A sends data to the AP.**
- 3. Identify which frame sets the NAV timer at STA-B and explain the role of the frame's Duration field.**

Solution:

1. STA-A and STA-B are hidden terminals because they cannot reliably hear one another.

When STA-A transmits, the signal arriving at STA-B is too weak to be detected as a busy medium. Consequently, STA-B's physical carrier sensing mechanism may determine that the channel is idle.

Therefore, STA-B may attempt to transmit even though STA-A is already communicating with the AP.

This creates the hidden-terminal problem.

2. RTS/CTS provides virtual carrier-sensing information that allows the hidden station to defer.

The exchange occurs as follows:

Step 1: STA-A first contends for the channel.

Step 2: STA-A sends an RTS frame to the AP.

Step 3: The AP receives the RTS and responds with a CTS frame after SIFS.

Step 4: STA-B may not hear the RTS because STA-A is outside its sensing range.

Step 5: STA-B can hear the CTS because STA-B is within range of the AP.

Step 6: STA-B reads the Duration information from the CTS and updates its NAV.

Step 7: STA-B defers transmission while the NAV indicates that the medium is reserved.

Step 8: STA-A transmits the Data frame to the AP without interference from STA-B.

3. The CTS frame received by STA-B sets or updates its NAV.

The Duration field specifies the remaining time required for the protected exchange, including the relevant data transmission and acknowledgement sequence.

STA-B uses this value to set its NAV timer. While $NAV > 0$, STA-B treats the medium as virtually busy and does not attempt normal channel access.

QUESTION 12 : An IEEE 802.11e station has two active Access Categories:

AC_VO and AC_BE

Both queues contain frames. At a particular time, the internal backoff entities for both Access Categories reach a backoff counter of zero simultaneously.

Questions:

- 1. Identify the term used to describe this condition.**
- 2. Explain how the station's internal scheduler resolves the conflict.**
- 3. Describe the post-collision actions taken by the winning and losing Access Categories.**

Solution:

1. This condition is called an Internal Collision, also referred to as a Virtual Collision.

An internal collision occurs when multiple Access Categories within the same station simultaneously obtain an opportunity to transmit.

However, a station has only one physical radio and therefore cannot transmit both frames simultaneously.

2. The internal scheduler compares the priorities of the competing Access Categories.

The standard EDCA priority ordering is:

$AC_VO > AC_VI > AC_BE > AC_BK$

Therefore, if AC_VO and AC_BE become eligible at the same time, AC_VO has the higher priority and is selected for transmission.

The winning Access Category is allowed to use the physical wireless medium.

3. The winning Access Category:

- Transmits its frame over the wireless medium.
- Obtains the corresponding EDCA Transmission Opportunity (TXOP), subject to its configured TXOP limit.

The losing Access Category:

- Does not transmit its frame over the wireless medium.

- Treats the internal collision as a collision for its contention state.
- Updates its retry state and contention window according to the applicable backoff procedure.
- Selects a new random backoff value before attempting channel access again.

Thus, an internal collision is resolved locally within the station, with the higher-priority Access Category winning the opportunity to transmit.